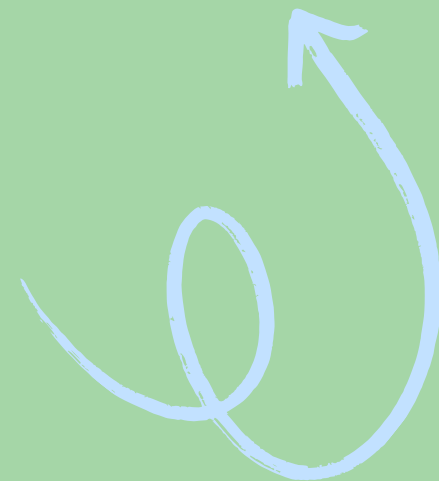


AI-Powered Crop Rotation Planner

Team Code: 022



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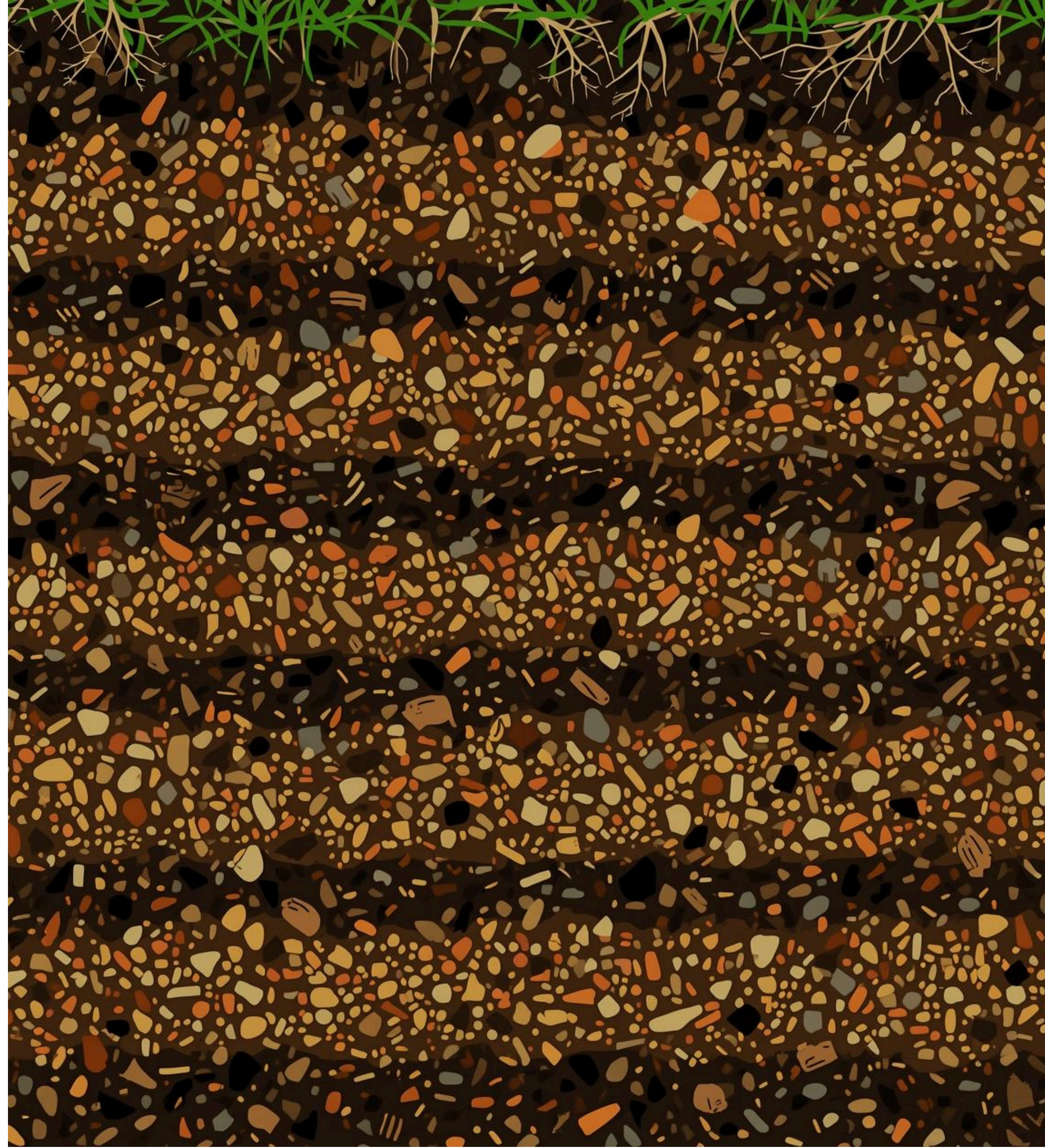
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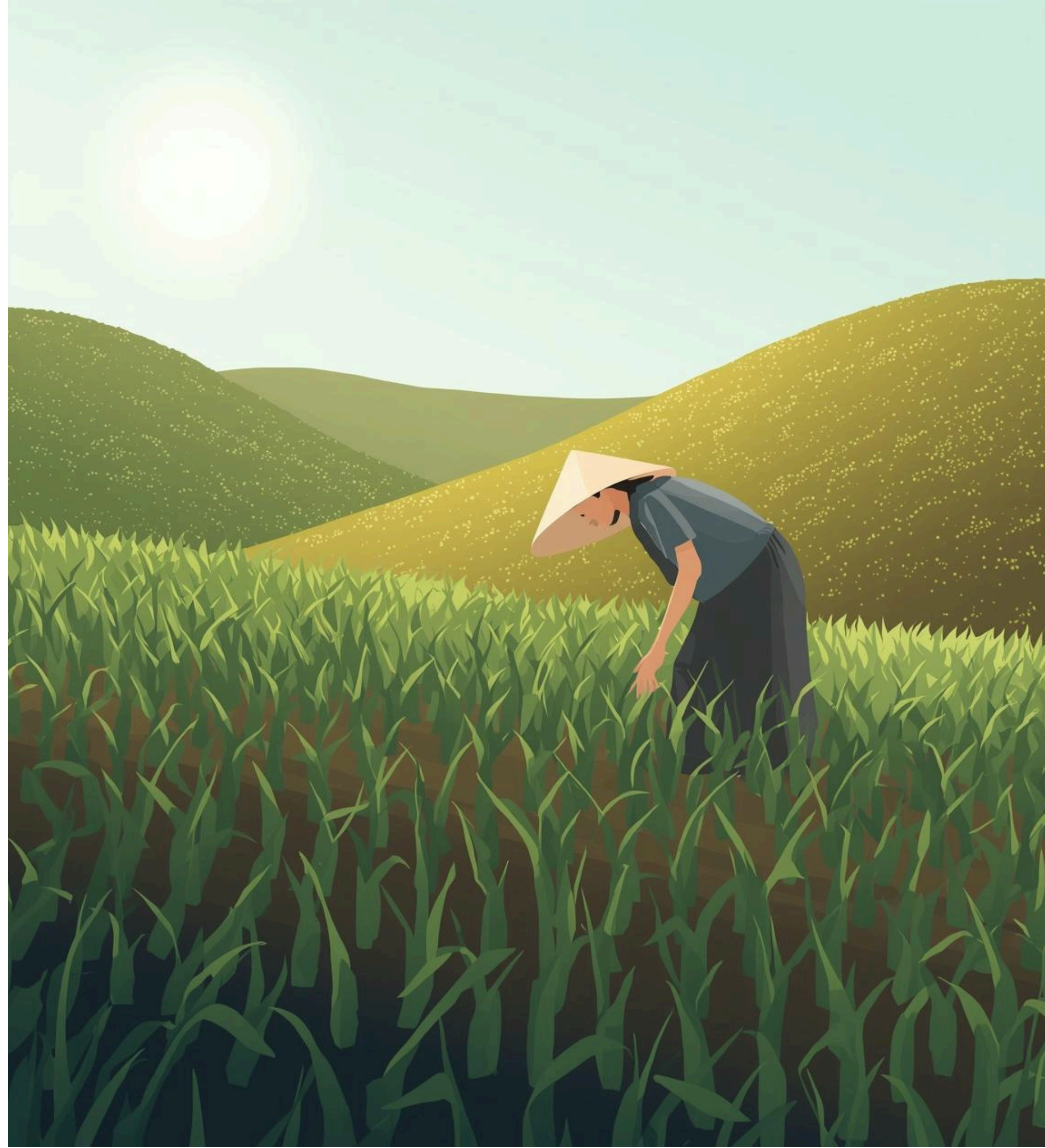
Current Agricultural Challenges

Soil nutrient depletion and increased pest attacks lead to reduced crop yield and higher fertilizer costs. Farmers often lack scientific guidance for planning crop sequences, resulting in inefficient practices.



Current Situation in Agriculture

Farmers predominantly depend on **traditional methods** for crop management, leading to inefficient planning. Limited digital solutions hinder effective crop rotation, leaving significant gaps in integrated companion planting guidance.



Our Solution

Unique Features for Sustainable Farming

Optimal Crop Suggestions

Our planner intelligently suggests the **best next crop** based on current conditions, ensuring effective use of resources while maintaining soil health and agricultural productivity.

Companion Crop Recommendations

The system provides tailored recommendations for **companion planting**, optimizing growth conditions and minimizing pest risks, enhancing the overall health of the crops cultivated together.

Rotational Planning

The planner generates a **comprehensive 2-3 year rotation plan**, promoting soil fertility and sustainability by integrating scientific principles and agricultural best practices for informed decision-making.

Tech Stack Overview

Essential components for our solution

Languages

Our application is built using **Python**, **HTML**, and **CSS**, ensuring a robust and flexible codebase that supports both backend and frontend development seamlessly.

Framework

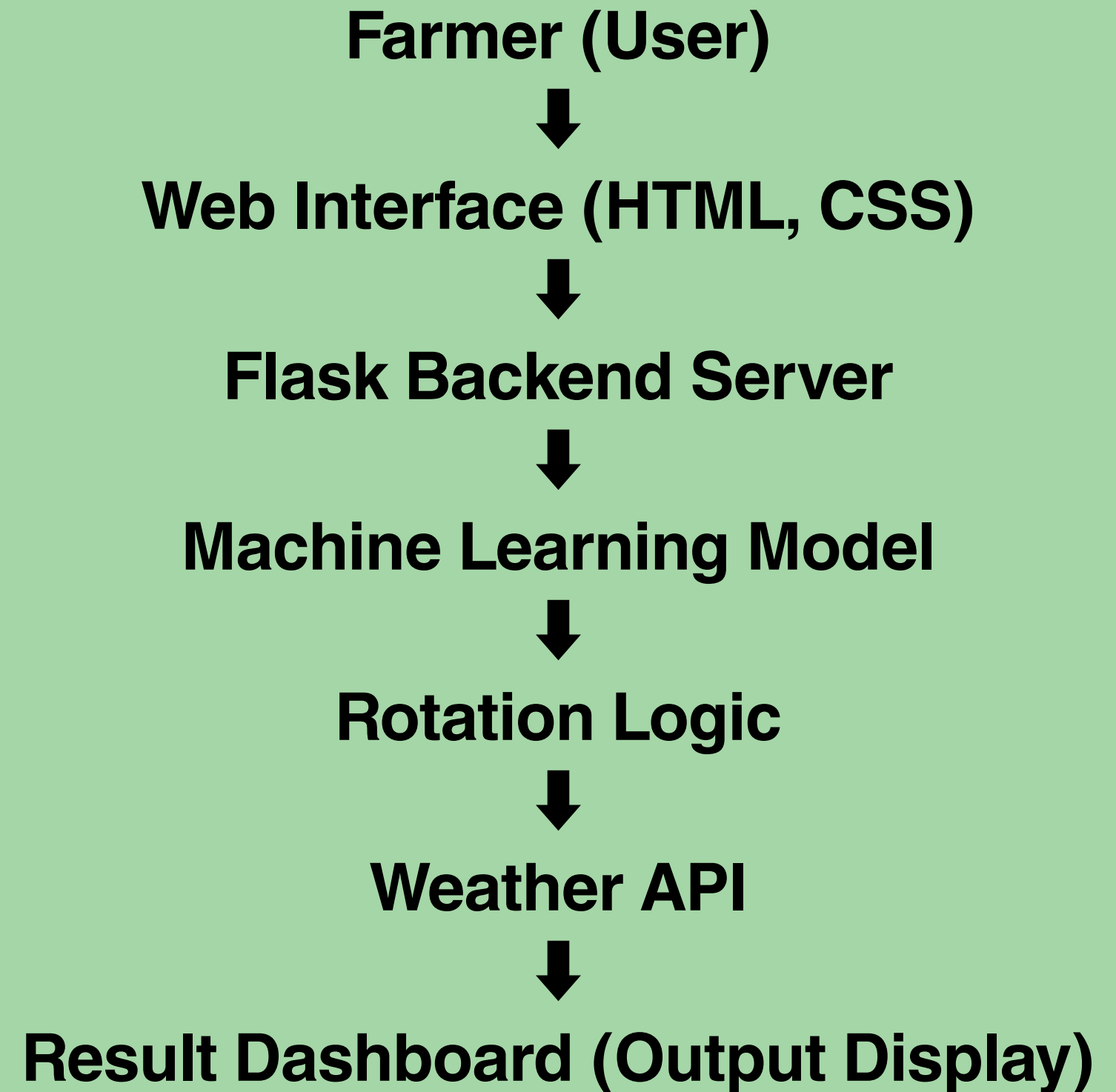
We utilize **Flask** as our framework, allowing for efficient handling of web requests and easy integration with our machine learning models for enhanced performance.

Libraries

Key libraries such as **Pandas** and **Scikit-learn** (if machine learning is implemented) provide the necessary tools for data manipulation and predictive analytics, crucial for our system's functionality.

System Architecture Overview

The architecture outlines the interaction flow from user input through the web interface, connecting to the Flask backend, which processes the rotation logic and machine learning model to display results effectively.



User Interaction

Steps for Using the System

Input Crop

The farmer begins by entering their **current crop** into the system, allowing the planner to assess the existing agricultural conditions and make informed recommendations.

Soil Type

Next, the farmer specifies the **soil type**, providing crucial data about nutrient composition and texture, essential for optimizing crop selection and rotation strategies for sustainable practices.

Seasonal Consideration

Finally, the system requires the **season** to tailor recommendations based on climatic conditions, ensuring that suggested crops align with the natural growing cycles and maximizing yield potential.

Project Phases

Key steps in development process

Phase 1

In this phase, we focus on **problem analysis and data collection** to identify specific agricultural challenges and gather relevant data for our planner.

Phase 2

During this phase, we will **design a rule-based or machine learning model** that will effectively analyze data and provide accurate recommendations for crop rotation and companion planting.

Phase 3

The third phase involves **frontend development**, where we create an intuitive user interface that allows farmers to easily input information and access the planner's recommendations.

Phase 4

Integrated the frontend with the Flask backend and connected all core modules.
Processed user inputs through rotation and companion planting logic.
Ensured smooth data flow and accurate recommendation generation.

Phase 5

Conducted end-to-end system testing for different crop and soil combinations.
Validated the accuracy of rotation and companion planting suggestions.
Fixed bugs and optimized overall performance and usability.

System Components

Key modules for effective planning

User Input Module

This module allows farmers to enter current crop information, soil type, and season, ensuring tailored recommendations based on specific conditions and requirements.

Crop Rotation Engine

The engine analyzes input data to suggest optimal crop rotations for improved soil health and yield, enhancing the overall farming strategy through data-driven decisions.

Companion Planting Engine

It identifies companion crops that benefit each other when planted together, promoting biodiversity and natural pest control while maximizing yield potential for farmers.

Impact of Our Solution

Key Benefits for Sustainable Agriculture

Improved Fertility

Our approach enhances soil fertility, promoting robust plant growth and enabling farmers to utilize natural resources more efficiently, ultimately leading to healthier ecosystems and sustainable yields.

Pest Reduction

By utilizing intelligent companion planting strategies, our solution effectively reduces pest attacks, minimizing reliance on chemical pesticides and fostering a more balanced ecosystem for crop growth.

Cost Efficiency

Implementing our planner significantly lowers fertilizer costs, as healthier soil naturally requires fewer external inputs, allowing farmers to allocate resources toward other essential agricultural practices.

Future Enhancements in Agriculture

Our planner will integrate **IoT soil sensors** for real-time data, a **Weather API** for accurate forecasting, and features like **market price analysis** and a **mobile app version** for greater accessibility.





Conclusion

Empowering farmers for sustainable growth